

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016–17

Zoology**Paper No. ZOB-503 : Biochemistry and Molecular Biology**

Time: 3 hours

Full Marks: 70

Note: Write your Roll No. at the top immediately on the receipt of this question paper. The figures in the right-hand margin indicate marks. Answer six questions including Question 1 which is compulsory. Coloured illustrations permitted.

Question 1. Answer the following parts:

(a) State whether the following statements are True or False with brief justification: $[8 \times 2 = 16]$

- (i) B-DNA and A-DNA are covalently closed circular DNA.
- (ii) CPSF inhibits capping of RNA.
- (iii) SeqA binds to RNA to initiate transcription.
- (iv) $(pK_1 + pK_2)/2$ is a common formulae to calculate pI of all types of α -amino acids.
- (v) A disulphide bond in a protein is generated between a cys and a met residue.
- (vi) Carnitine shuttle aids in oxidative phosphorylation.
- (vii) A sigmoidal kinetics is shown by a multimeric enzyme.
- (viii) Cerebrosides are derivatives of sphingolipids.

(b) Match each item from Column–X with one from Column–Y: $[4 \times \frac{1}{2} = 2]$

Column–X		Column–Y	
(i)	Palindrome	(1)	Tryptophan
(ii)	UAA	(2)	Restriction endonucleases
(iii)	UGA	(3)	Termination
(iv)	GUG	(4)	Initiation of translation
		(5)	RNA polymerase-I

(c) Define the following: $[2 \times 1 = 2]$

- (i) Codon
- (ii) Protein domain.

Question 2. Answer the following parts:

- (a) Represent steps of one cycle of Edman's reaction with a polypeptide chain. [5]
- (b) Using Lineweaver-Burk equation, justify all the kinetic parameters shown in the LB plot. [5]

Question 3. Answer the following parts:

- (a) Compare stabilization of α -helix and β -pleated sheet structures of polypeptides. [6]
- (b) Write pay-off reactions of the glycolytic pathway. [4]

Question 4. Answer the following parts:

- (a) Explain chemiosmotic concept of ATP synthesis. [5]
- (b) Write de-carboxylation reactions of the TCA cycle. [5]

Question 5. With the help of well-labeled diagrams, explain the following: [2 × 5 = 10]

- (a) Formation of pre-initiation complex at promoter for RNA Pol II
- (b) Mechanism of termination of translation in prokaryotes.

Question 6. Describe compositions and functions of the following: [2 × 5 = 10]

- (a) Telomerase
- (b) Nucleosome.

Question 7. Differentiate between the following: [2 × 5 = 10]

- (a) Cloning and Expression vectors
- (b) Primosome and Replisome.

Question 8. Write short notes on the following: [2 × 5 = 10]

- (a) Phospholipids
- (b) Ribosome.

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